

Week - 05 - Classes, Graphics\Lecture_9_code_3.py

```

1 import math
2 import time
3
4 G = 9.82
5 AIR_DRAG = 0.4
6
7 class Projectile:
8     def __init__(self, ID, size, energy, x_pos=0, y_pos=0):
9         self.id = ID
10        self.size = size
11        self.energy = energy
12        self.x_pos = x_pos
13        self.y_pos = y_pos
14
15    def get_projectile_data(self):
16        # A getter function
17        return (f'id {self.id}, size {self.size}, energy {self.energy} and position
18                ({self.x_pos}, {self.y_pos})')
19
20    def get_bomb_ids(self):
21        return self.id
22
23    def fire(self, velocity, theta):
24        self.x_pos
25        self.y_pos
26        dt = 0.02
27
28        x_vel = velocity * math.cos(theta)
29        y_vel = velocity * math.sin(theta)
30
31        while self.y_pos >= 0:
32            self.x_pos += x_vel * dt
33            self.y_pos += y_vel * dt
34            x_vel -= AIR_DRAG * dt
35            y_vel -= G * dt
36
37            time.sleep(0.1)
38            animate(self.x_pos, self.y_pos)
39
40 # Helper function to display trajectory of our projectile
41 def animate(x, y):
42     width, height = 60, 30
43     scale = 100
44     screen = [[" " for _ in range(width)] for _ in range(height)]
45
46     column = round(x * scale)
47     row = height - 1 - round(y * scale)
48     if 0 <= column < width and 0 <= row < height:
49         screen[row][column] = "*"
50
51     print("\033[2J\033[H", end="")
52     print("\n".join("".join(line) for line in screen), flush=True)

```

```
52     time.sleep(0.1)
53
54
55 b1 = Projectile(201, 10, 3.65)
56 print(b1.get_projectile_data())
57 # b1.fire(20, math.radians(45)) velocity 20 was pushing projectile out of view in lecture
58 b1.fire(2, math.radians(45))    # shift to velocity 2
```